

**The Hong Kong Polytechnic University**

**Department of Electronic and Information Engineering**

**EIE2111 Lab 3: Control Statements (Part 1)**

**(Deadline for Submission: Check the course information)**

**Introduction**

This laboratory exercise is designed to give you hand-on experience in Control Statements (Part 1).

**Questions**

1. One large chemical company pays its salespeople on a commission basis. The salespeople each receives \$200 per week plus 9 percent of their gross sales for that week. For example, a salesperson who sells \$5000 worth of chemicals in a week receives \$200 plus 9 percent of \$5000, or a total of \$650. Develop a C++ program that uses a while statement to input each salesperson's gross sales for last week and calculates and displays that salesperson's earnings. You should process one salesperson's figure at a time. If the input is -1, the program should exit.
2. Write a program that reads the side of a square and then prints a hollow square of that size out of asterisks and blanks. Your program should work for squares of all sides between 1 and 20.
3. A company wants to transmit data over the telephone, but is concerned that its phones could be tapped. All of the data are transmitted as four-digit integers. The company has asked you to write a program that encrypts the data so that it can be transmitted more securely. Your program should read a four-digit integer and encrypt it as follows: Replace each digit by *(the sum of that digit plus 7) modulus 10*. Then, swap the first digit with the third, swap the second digit with the fourth and print the encrypted integer. Write a separate program that inputs an encrypted four-digit integer and decrypts it to form the original number.
4. The factorial of a non-negative integer  $n$  is written  $n!$  and is defined as follows:

$$n! = \begin{cases} 1 & \text{if } n = 0 \text{ or } 1 \\ n \times (n-1) \times (n-2) \times \dots \times 2 \times 1 & \text{otherwise} \end{cases}$$

- a) Write a program that reads a non-negative integer, and computes and prints its factorial.
- b) Write a program that estimates the value of the Mathematical constant  $e$  by using the formula

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots$$

Prompt the user for the desired accuracy (i.e., the number of terms in the summation).

c) Write a program that computes the value of  $e^x$  by using the formula

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Prompt the user for the desired accuracy (i.e., the number of terms in the summation).

5. Combine the above four programs into one. The following is the sample output:

```
C:\Windows\system32\cmd.exe
Question 1
Enter sales in dollars <-1 to end>: 500
Salary is: $245
Enter sales in dollars <-1 to end>: 65
Salary is: $205.85
Enter sales in dollars <-1 to end>: -1

Question 2
Enter length of side: 6
*****
*   *
*   *
*   *
*   *
*****

Question 3
Enter a four-digit number: 1234
Encrypted number is 0189
Enter an encrypted number: 1234
Decrypted number is 6745

Question 4
Enter a positive Integer: 4
4! is 24
Enter desired accuracy: 3
e is 2.5
Enter exponent x: 4
Enter desired accuracy: 6
e^x = 42.8667

Press any key to continue . . .
```

## Instructions

- You are required to submit your C++ programs (the whole projects created in Microsoft Visual Studio 2019) in Question 5 to Blackboard. Zip all of them into a single file.
- The deadline of the submission: **Check the course information.**
- It is not required to create any classes or header files for the above exercises. It is fine if all program codes are in the main program.

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